

Coding & Solution

Date	3 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/AmruthaImmidisetti/Smart-Lender.git
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn, XGBoost, Pandas, NumPy
Key Features Implemented	Loan eligibility prediction, applicant data preprocessing, XGBoost model integration, real-time prediction, user-friendly Flask web interface, input validation.
Pending / Incomplete Features	IBM Cloud deployment optimization, user authentication, prediction history, and admin dashboard. <i>(If everything is complete, write None.)</i>
Setup / Run Instructions	Install dependencies using <code>pip install -r requirements.txt</code> , run the Flask application using <code>python app.py</code> , then open the provided localhost URL in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The Smart Lender application follows a modular architecture with separate components for data preprocessing, model training, prediction, and the Flask web interface. XGBoost was selected as the final model based on its superior performance, achieving **94.7% training accuracy** and **81.1% testing accuracy**. The codebase is well-structured, maintainable, and designed for future enhancements and cloud deployment.